Program 3: Working with Selenium: Maven Project To Connect To Web Servers Using Selenium Dependency

1.Sauce.demo webpage

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
samiksha@samiksha:~/report$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=mav
en-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/samiksha/report
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/report/MyMavenSeleniumApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.456 s
[INFO] Finished at: 2026-03-17T20:40:07+05:30
[INFO] ------------------------------------------------------------------------
```

Step 2: Before we connect to web servers we need to have chrome driver installed. To download chrome driver

To check the chrome version type the command **google-chrome –version** in the terminal

```
samiksha@samiksha:~/report$ google-chrome --version
Google Chrome 145.0.7632.116
samiksha@samiksha:~/report$
```

extract the version and store it in a variable CHROME_VERSION

```
samiksha@samiksha:~/report$ CHROME_VERSION=$(google-chrome --version | awk '{print $3}'| cut -d'.' -f1-3)
samiksha@samiksha:~/report$
```

extracts and transorms JSON data

LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latest-patch-versions-per-build.json" | jq -r ".builds[\"$CHROME_VERSION\"].version")

do install curl using the command

sudo apt install curl

wget https://storage.googleapis.com/chrome-for-testing-public/$LATEST_DRIVER/linux64/chromedriver-linux64.zip

Extract and move it to /usr/local/bin/:

unzip chromedriver-linux64.zip

sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver
sudo chmod +x /usr/local/bin/chromedriver
Check if ChromeDriver is installed correctly:

```
samiksha@samiksha:~/report$ chromedriver --version
ChromeDriver 145.0.7632.117 (de4e5d6e13b995f5e04137970c0e8ec456017467-refs/branch-heads/7632@{#3101})
samiksha@samiksha:~/report$
```

Step 3 : edit the pom.xml file and add the dependencies
The pom.xml has the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp01</artifactId>
  <packaging>jar</packagi<ng>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp01</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.29.0</version>
</dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
```

```
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>

        </plugin>

</plugins>
</build>
</project>
```

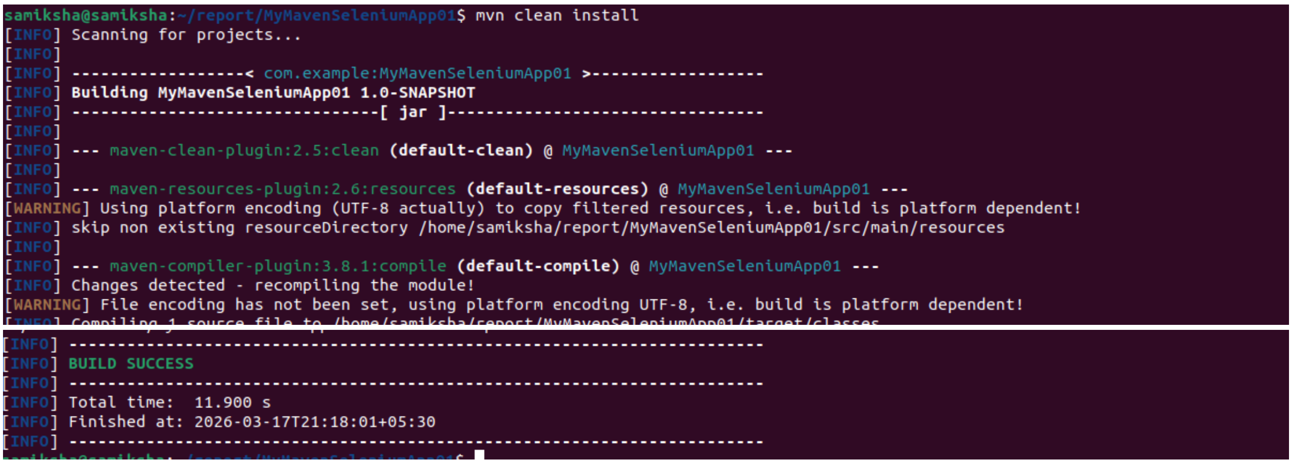
Step 4: Edit the App.java

```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

/**
 * Hello world!
 *
 */
public class App
{
    public static void main( String[] args )
    {
        WebDriver driver=new ChromeDriver();
        driver.get("https://www.saucedemo.com/");
        driver.manage().window().maximize();
```

```
        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        driver.findElement(By.id("login-button")).click();
    }
}
```

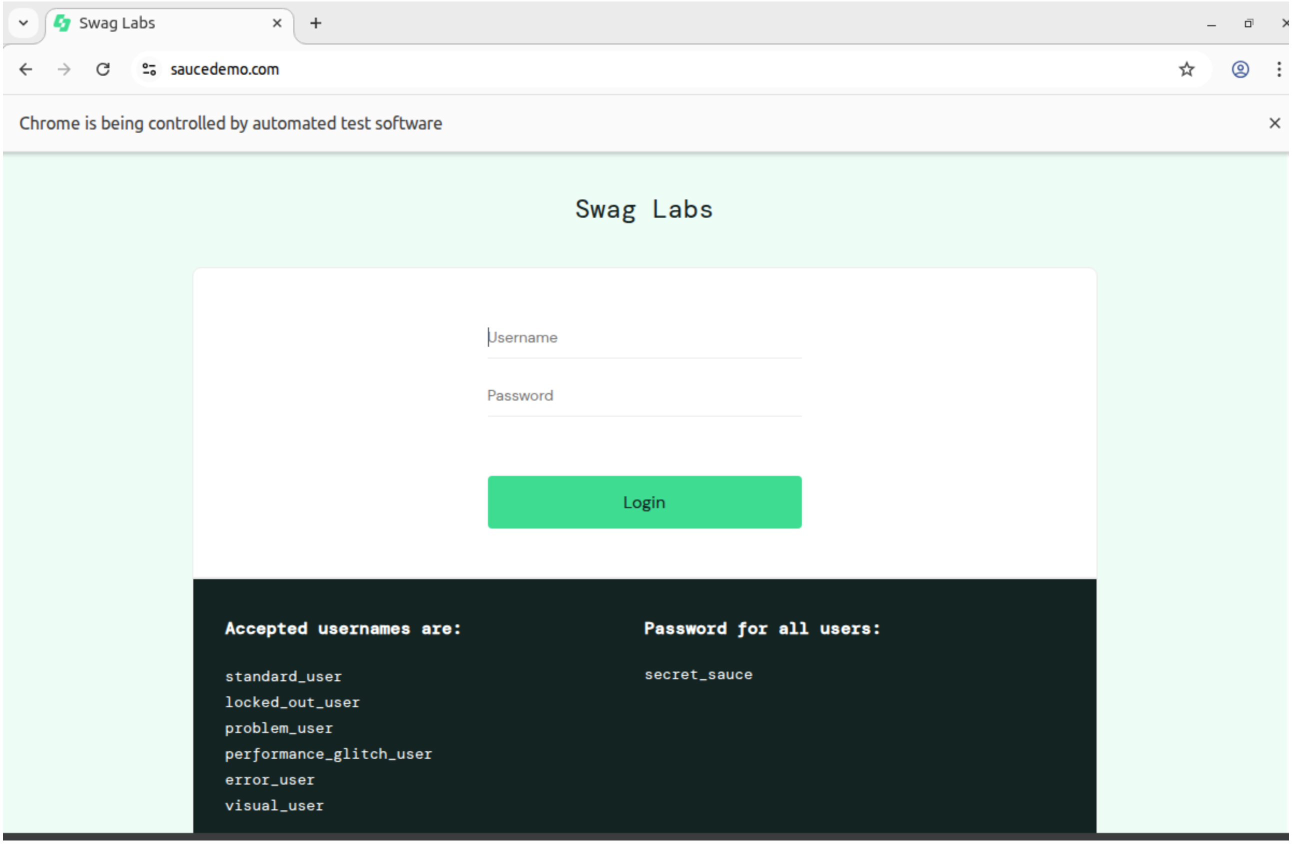
Step 5:build and run the maven application

Type **mvn clean install** in the terminal



To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"**

it opens saucedemo.com and we can login using the credentials



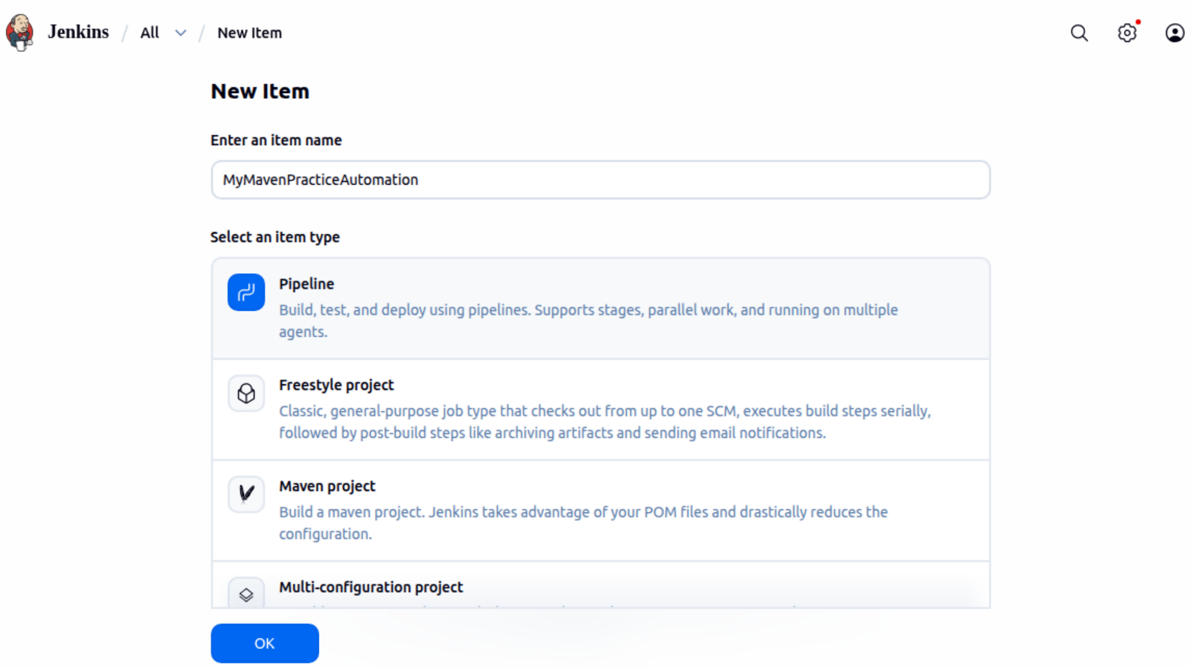
Step 6: now open the terminal and type the command **cd MyMavenPracticeAutomationApp** to enter the folder and create a **Jenkinsfile** .Paste the code in the Jenkinsfile

```
pipeline {
    agent any  // Use any available agent
    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/Samiksha201104/MyMavenPracticeAutomationApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
```

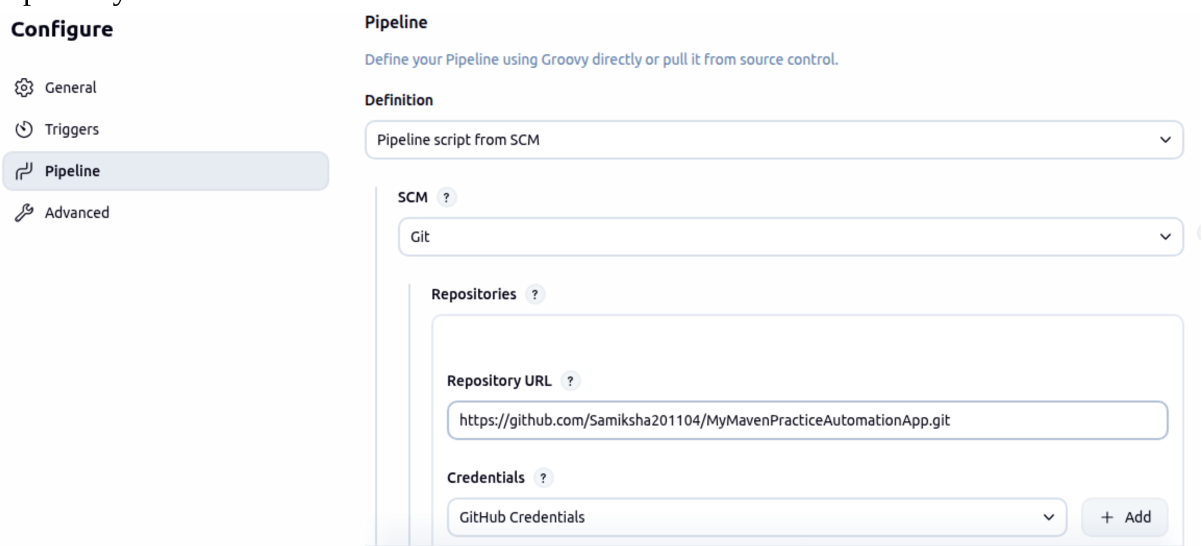
```
        }
        failure {
            echo 'Build failed!'
        }}}
```

Then commit and push it to GitHub

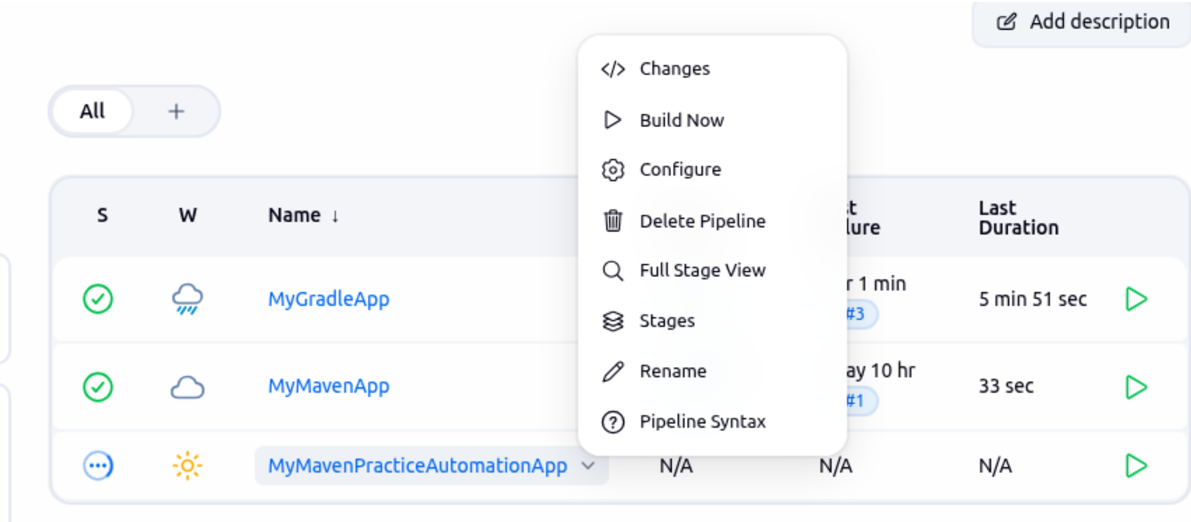
Step 7:in the Jenkins Dashboard click on new item type the name of the item and select Pipeline

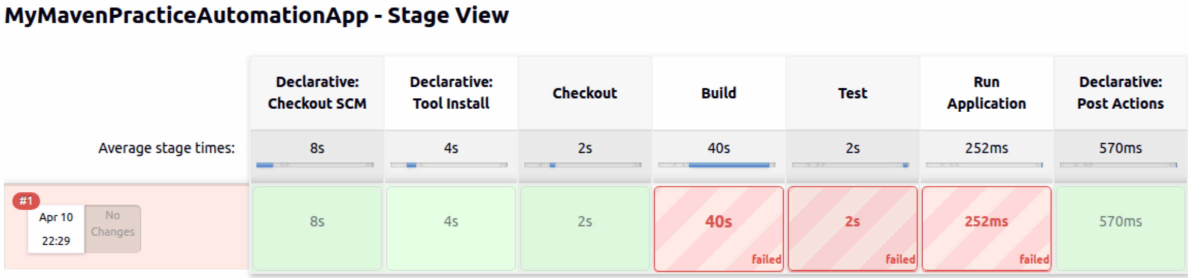


In configure scroll down to Pipeline and choose pipeline script from SCM,give the repository url and choose GitHub Credentials



Step 8 : in the Dashboard click on the MyMavenPracticeAutomationApp item and select Build Now and then Full Stage view





To correct the errors change the code in the pom.xml

<source>1.7</source>
<target>1.7</target>

To

<source>17</source>
<target>17</target>

Save and Build again in Jenkins